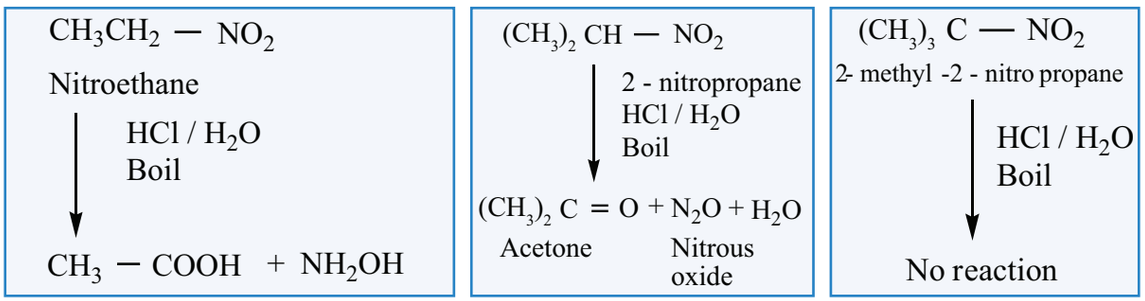


On the other hand, the acid or base hydrolysis of ethyl nitrite gives ethanol.

$$\underset{\text{Ethylnitrite}}{CH_3CH_2-O-N=O} + HOH \xrightarrow[(\text{or})\ H^+]{OH^-} \underset{\text{Ethanol}}{CH_3CH_2-OH} + HNO_2$$

iii. Halogenation of nitroalkanes

Primary and secondary nitroalkanes on treatement with Cl_2 or Br_2 in the presence of NaOH give halonitroalkanes. The α - H atom of nitroalkanes are successively replaced by halogen atoms.

$$CH_3 - NO_2 + 3Cl_2 \xrightarrow{NaOH} \underset{\text{Chloropicrin (trichloronitromethane)}}{CCl_3 - NO_2} + 3HCl$$

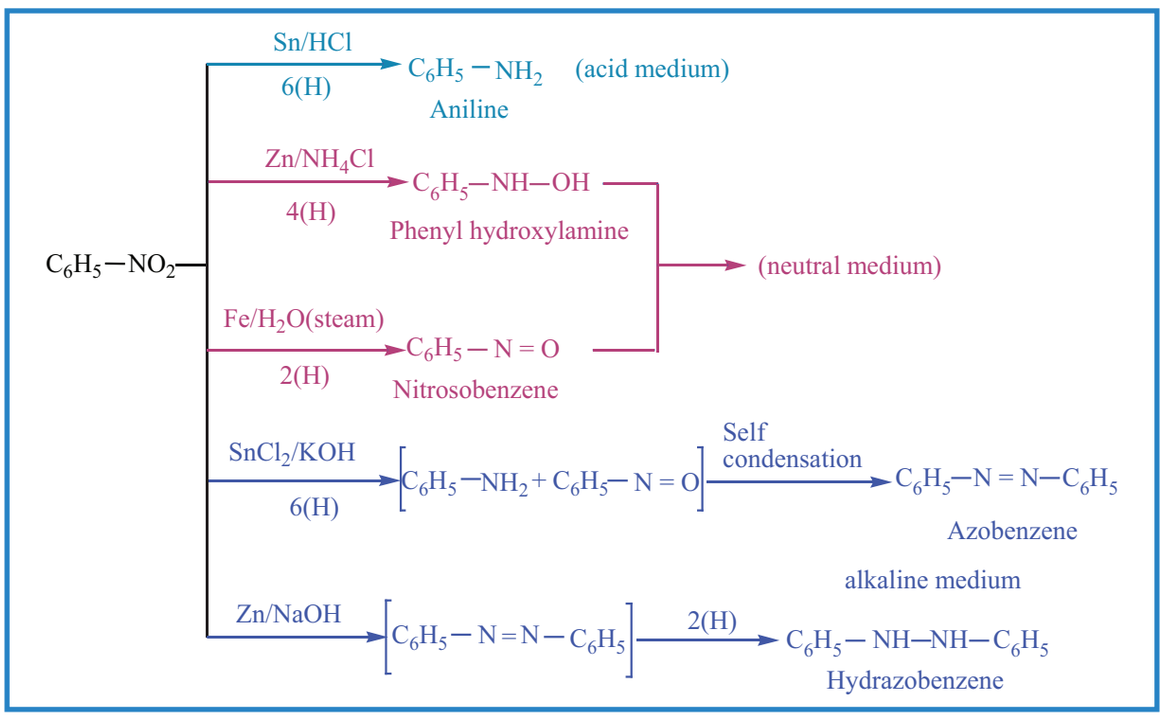
Toxicity

Nitroethane is suspected to cause genetic damage and be harmful to the nervous system.

iv. Nef carbonyl synthesis :

$$CH_3CH_2NO_2 \xrightarrow{KOH} CH_3-CH=N\begin{matrix}O^{\ominus}\\ \diagup\\ \diagdown\\ OH\end{matrix} \xrightarrow{H_2O/H^+} \underset{\text{Nitroxyl azanone}}{CH_3-CHO} + HNO$$

Chemical Properties of nitrobenzene



XII U13-Organic Nitrogen Compounds.indd 203 16-12-2022 19:45:09